

OCCURRENCE OF PRIMIDONE, CARBAMAZEPINE, CAFFEINE, AND PRECURSORS FOR *N*-NITROSODIMETHYLAMINE IN DRINKING WATER SOURCES IMPACTED BY WASTEWATER¹

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ABSTRACT: Wastewater impact on drinking water sources was assessed using several approaches, including analysis of three pharmaceuticals and personal care products (PPCPs) – primidone, carbamazepine, and caffeine – as indicators, and determination of precursor concentrations for the disinfection byproduct *N*-nitrosodimethylamine (NDMA) using formation potential (FP) tests. Samples were collected in 2006 and 2007 in rivers impacted by wastewater treatment plant (WWTP) discharges, at drinking water treatment plant (DWTP) intakes upstream or downstream from these discharges, and from two WWTP effluents in two watersheds. The levels [10th percentile – maximum (median)] of primidone, carbamazepine, caffeine, and NDMAFP were 2-95 (7) ng/l, 2-207 (18) ng/l, 7-687 (78) ng/l, and 12-321 (35) ng/l, respectively. The highest concentrations of primidone, carbamazepine, and NDMA precursors were from one of the WWTP effluents, whereas the highest concentration of caffeine was detected in a river heavily impacted by treated wastewater discharges. The lowest concentrations of the three PPCPs were from a DWTP influent upstream of a metropolitan urban area with minimum wastewater impact. Temporal variations in PPCP and NDMAFP concentrations and streamflows in two selected watersheds were also observed. Furthermore, correlation analysis between caffeine or carbamazepine and primidone was evaluated. The results show that measurement of the two pharmaceuticals and NDMAFP tests can be used to evaluate wastewater impact in different watersheds, whereas caffeine results were more variable.

(KEY TERMS: pharmaceuticals and personal care products; primidone; carbamazepine; caffeine; disinfection byproducts; *N*-nitrosodimethylamine; formation potential; wastewater indicators; drinking water; wastewater impact.)

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INTRODUCTION

Wastewater treatment plant (WWTP) discharges are one of the main sources of various micropollutants in the environment (Kolpin *et al.*, 2002; Snyder *et al.*, 2003), including pharmaceuticals and personal care products (PPCPs), and certain disinfection byproducts

(DBPs) and their precursors (Krasner *et al.*, 2008). The occurrence of PPCPs in surface waters affected by treated wastewater discharges has been reported by the U.S. Geological Survey and other research groups (Kolpin *et al.*, 2002; Glassmeyer *et al.*, 2005). In addition, PPCPs have been used as potential indicators of human fecal contamination (Glassmeyer *et al.*, 2005). Previously, some researchers used boron as a

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wastewater indicator (Schreiber and Mitch, 2006). However, in many waters in the western United States (U.S.), ambient levels of boron in such waters preclude the use of this chemical as a wastewater indicator. As part of a project on the occurrence and formation of nitrogenous DBPs (N-DBPs), wastewater impact on drinking water sources was assessed using several approaches (W.A. Mitch *et al.*, 2008), including analysis of three PPCPs – primidone, carbamazepine, and caffeine – as indicators, and determination of precursor concentrations for the DBP *N*-nitrosodimethylamine (NDMA). The anticonvulsants primidone and carbamazepine are stable in the environment and are considered conservative indicators (Sedlak *et al.*, 2004; Glassmeyer *et al.*, 2005; Krasner *et al.*, 2006). Caffeine has been identified as an anthropogenic marker for wastewater contamination of surface waters; however, it can undergo biodegradation (Buerge *et al.*, 2003). All three PPCPs have been detected in treated wastewater (Glassmeyer *et al.*, 2005; Krasner *et al.*, 2006) at levels considerably higher than the analytical method's minimum reporting levels (MRLs); therefore their fate, transport, and attenuation in watersheds can be followed. Treated wastewater discharges are also a source for NDMA precursors (Mitch and Sedlak, 2004). An NDMA formation potential (FP) test under reactivity-based chloramination conditions (Krasner *et al.*, 2004) has been shown as a useful tool to evaluate NDMA precursor levels and to better understand the impact of treated wastewater discharges on downstream drinking water sources (Krasner *et al.*, 2008).

ANALYTICAL METHODS

Materials

Primidone, carbamazepine, caffeine, and NDMA were obtained from Sigma-Aldrich (St. Louis, Missouri). Carbamazepine- d_{10} , caffeine- $^{13}C_3$, and NDMA- d_6 were obtained from Cambridge Isotope Laboratories (Andover, Massachusetts). Primidone- d_5 was obtained from C/D/N Isotopes Inc. (Quebec, Canada). OmniSolv grade methanol (MeOH) was purchased from VWR (West Chester, Pennsylvania). Organic-pure water (OPW) was obtained from a Millipore UV Plus system (Billerica, Massachusetts).

Sample Collection

The authors collected 12 samples in 2006 from five rivers and seven drinking water treatment plant

(DWTP) intakes, upstream or downstream from wastewater discharges. The authors also sampled one WWTP effluent in 2006. The sample sites were geographically distributed, covering the west, mountain, south central, and northeast regions in the U.S. In 2007, the authors collected a total of 16 samples (15 samples from 3 rivers and 12 DWTP intakes and an additional WWTP effluent). In addition, the re-sampling in certain watersheds provided information on temporal variations. The sampling locations represented a range of distances away from WWTP discharge points, from a few miles to stretches across an entire state. The mixing conditions of the watersheds sampled varied from poorly mixed to well mixed. However, obtaining detailed hydrological information on the water bodies in this study was outside the scope of work of the project.

Samples were collected in 1-l amber glass bottles and shipped to the laboratory overnight in ice chests with frozen Blue Ice. All samples were grab samples. Previously, the authors found that the concentrations of primidone did not vary significantly at one WWTP within a 24-h period based on a diurnal study (Krasner *et al.*, 2006). A Lagrangian sampling plan to follow the same parcel of water as it moves through each watershed was not used for this study, therefore the samples collected along a stretch of river were not collected at times that matched the actual flow times. However, in a previous study, primidone samples were collected from an effluent-dominated river in two sample events, either once each day on consecutive days or twice on the same day, and the values were not found to be significantly different over these time frames (Krasner *et al.*, 2006). Upon arrival at the laboratory, samples were filtered with nylon membrane filters (0.45 μm) with a glass microfiber prefilter, acidified to pH 2 with 4 M hydrochloric acid (Vanderford *et al.*, 2003), and kept at 4°C until extraction. To evaluate sample collection protocols and potential contamination issues, field blanks were collected at selected sample sites, where OPW was poured into empty sample bottles during the same time period when samples were collected. The field blanks were processed in the same way as the environmental samples.

Sample Analysis

Solid-phase extraction (SPE) of the PPCPs was performed on an AutoTrace automated SPE work station (Caliper Life Sciences, Hopkinton, Massachusetts), using 200-mg hydrophilic-lipophilic-balance cartridges from Waters Corporation (Milford, Massachusetts). The cartridges were conditioned with 5 ml of MeOH followed by 5 ml of OPW. Samples (500 ml)

were loaded onto the cartridges, after which the cartridges were rinsed with 5 ml of OPW and dried with nitrogen for 60 min. The cartridges were eluted with 8 ml of MeOH, which was concentrated down to 0.5 ml on a TurboVap evaporation system (Caliper Life Sciences). Immediately before analysis, 0.5 ml of OPW was added to the sample extract.

The sample extracts were analyzed by liquid chromatography/tandem mass spectrometry (LC/MS/MS) under electrospray positive ionization mode (Vanderford *et al.*, 2003) on an Applied Biosystems (Foster City, California) API 4000 triple quadrupole MS coupled with an Agilent (Santa Clara, California) 1100 LC. LC/MS has been increasingly used in the analysis of PPCPs in water as a result of its ability to analyze polar, thermally labile, and nonvolatile chemicals, which are not amenable to the traditional gas chromatography (GC)/MS analysis without extra derivatization steps (Vanderford *et al.*, 2003). With recent advancement in MS instrumentation, PPCPs can be detected in the low ng/l range in water, when SPE (or other extraction techniques) is used (Vanderford *et al.*, 2003; Vanderford and Snyder, 2006). To increase selectivity, MS/MS is used in PPCP analysis by monitoring the transition from a selected precursor ion (generally a protonated molecular ion for electrospray positive ionization) to a product ion, minimizing interference from the matrix. The analytes for this study were eluted off a 50 × 2 mm Phenomenex (Torrance, California) Synagi Max-RP C12 column with 4-μm particle size, using a binary gradient of 0.1% formic acid (v/v) in 95% water: 5% MeOH (A) and MeOH (B). The elution gradient was as follows: held at 100% A for 1 min, increased to 90% B by 4 min, held for 1 min, increased to 100% B by 6 min, held for 1 min. The initial equilibration time was 4 min. The instrument operating conditions are listed in Table 1.

The precursor/product ion pairs used for the MS/MS transitions are listed in Table 2, together with retention times for the analytes (W.A. Mitch *et al.*,

TABLE 1. Instrument Operating Conditions.

LC Operating Conditions	
Flow rate	0.4 ml/min
Autosampler temperature	10°C
Injection volume	20 μl
Column temperature	Ambient
MS/MS Operating Conditions	
Collision gas	6 psi
Curtain gas	10 psi
Ion spray voltage	5,000 V
Temperature	500°C
Declustering potential	70 V
Entrance potential	10 V
Collision energy	25 eV
Collision cell exit potential	10 V

Notes: LC, liquid chromatography; MS, mass spectrometry.

TABLE 2. Precursor/Product Ion Pairs Used in the Analysis.

Compound	Precursor Ion	Product Ion	Retention Time (min)
Primidone	219	162	4.85
Primidone-d ₅	224	167	4.83
Carbamazepine	237	194	5.46
Carbamazepine-d ₁₀	247	204	5.44
Caffeine	195	138	4.42
Caffeine- ³ C ₁₃	198	140	4.42

Source: Reproduced from W.A. Mitch *et al.* (2008). Copyright Awwa Research Foundation.

2008). Compounds were identified by matching both the retention time and MS/MS transition of each analyte in the samples with those of the same analyte in authentic standards. Representative LC/MS/MS total ion chromatograms of the three analytes together with the three isotopically labeled internal standards in an extracted standard and in a surface water sample are shown in Figure 1.

FP tests for NDMA were conducted according to the protocols developed during a previous study (Krasner *et al.*, 2004). The samples were analyzed by GC/MS under chemical ionization mode (Guo *et al.*, 2004; Cheng *et al.*, 2006).

Method Detection Limits and Calibration

The method detection limit (MDL) for each analyte was determined in accordance with U.S. Environmental Protection Agency (USEPA) guidelines (USEPA, 1990). A set of seven replicate samples fortified at or near the expected MDLs were extracted and analyzed according to the standard operating procedures. The MDL of each analyte was calculated using the following formula: MDL = 3.14 × SD, where the coefficient 3.14 represents the Student's *t*-value for *n* – 1 degrees of freedom. For the purpose of this study, MRLs were defined as three times the corresponding MDLs, and were determined to be 1 ng/l for each of the three PPCPs and 10 ng/l for NDMAFP. Standard curves generally contained a minimum of four points, ranging from the analyte's respective MRL to 250 ng/l. Isotopically labeled standards were added to both calibration standards and samples at 20 ng/l each. On occasions when the concentration of an analyte in a particular sample exceeded the highest point of calibration, the sample was diluted and re-analyzed.

Quality Control

Quality control protocols included method blanks, duplicate samples, laboratory fortified (spiked)

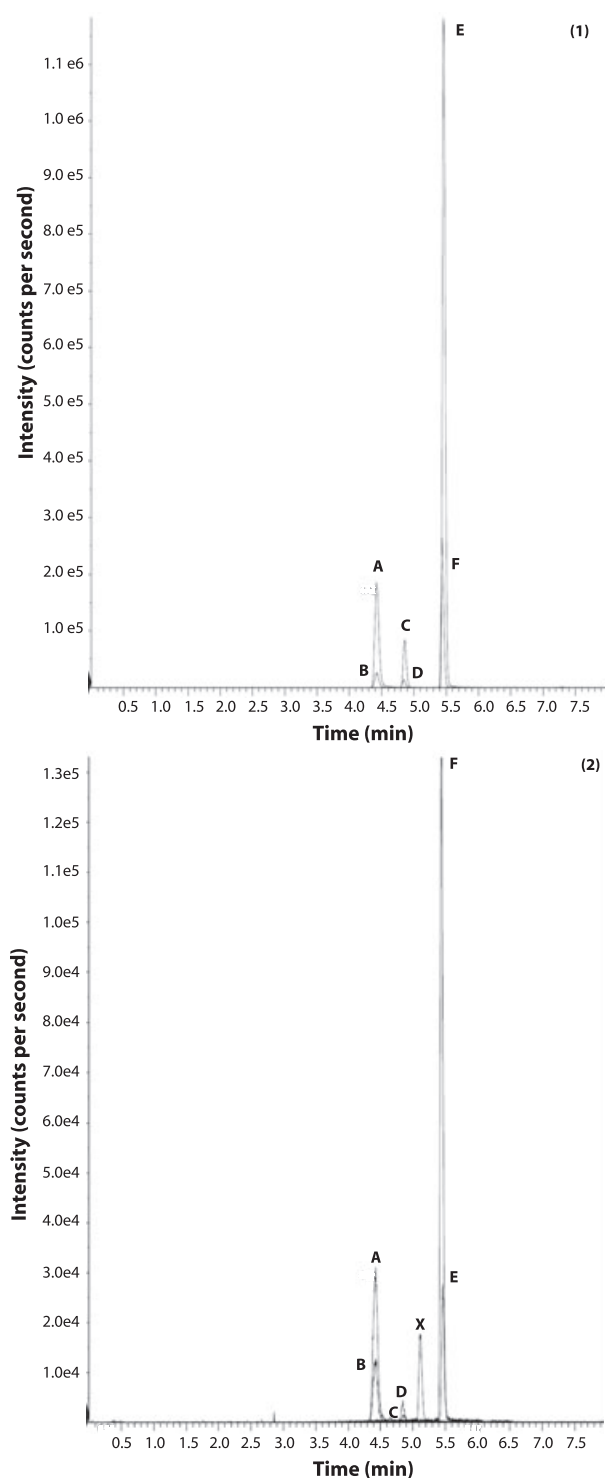


FIGURE 1. Total Ion Chromatograms of (1) an Extracted Standard With 100 ng/l of Each Analyte and 20 ng/l of Each Labeled Standard and (2) a Surface Water Sample With 33 ng/l of Caffeine, 6 ng/l of Primidone, 3 ng/l of Carbamazepine, and 20 ng/l of Each Labeled Standard. Keys: (A) caffeine, (B) caffeine-³C₁₃, (C) primidone, (D) primidone-d₅, (E) carbamazepine, (F) carbamazepine-d₁₀, and (X) unknown peak in the sample only.

samples, and selected field blanks. For each batch of 12 samples, at a minimum, one laboratory blank, one duplicate sample, and one spiked sample were analyzed. The laboratory blanks had no detectable amount of the three PPCPs, but in a few occasions contained detectable amounts of NDMAFP, where the NDMAFP in the blanks was less than half of its MRL. Of the 13 samples collected in 2006, three field blanks were collected and analyzed. Carbamazepine was found in one field blank at 1 ng/l, and caffeine was found in two of the three field blanks at 1-2 ng/l. These levels were 10 or more times lower than the amounts in the samples from the corresponding sites. The relative percent differences of duplicate samples were all within 20%, which indicated that the precision of the analytical methods was acceptable to enable comparison of results from different sites. When comparing the spiked samples *vs.* the unspiked counterparts, the percent recoveries of the spiked amounts were from 70% to 130% for primidone, carbamazepine, caffeine, and NDMAFP, with the exception of the primidone results in 2006, which are further discussed below.

One of the drawbacks of LC/MS analysis, especially in the electrospray mode, is that matrix suppression and enhancement often occurs, which adversely affect the accuracy of the results. The most effective way to compensate for matrix effects has been reported to be isotope dilution, where an isotopically labeled standard is used as the internal standard for the unlabeled counterpart (Vanderford and Snyder, 2006). This isotope dilution approach was used for carbamazepine and caffeine throughout the project, but isotopically labeled primidone was not commercially available when samples were collected in 2006, therefore caffeine-³C₁₃ was used as the internal standard for primidone. As a result, the spike recovery for primidone varied from 20% to 70%, depending on the matrix. To compensate for the variable recoveries, all but two samples collected in 2006 were run twice, once as the native sample and the other matrix fortified, and the results in the unspiked sample were corrected using the spike recovery. In addition, selected samples were analyzed for primidone by a derivatization-GC/MS method (Krasner *et al.*, 2006) to confirm the quantitation. In 2007, primidone-d₅ was custom synthesized and was used in sample analysis as the internal standard for primidone. This practice greatly improved the spike recoveries of primidone to the same 70-130% range as the other analytes, and the primidone results were not adjusted for spike recovery in 2007. These spikes recoveries indicate that the accuracy of the analytical methods was acceptable for making comparisons between sites.

Statistical Analysis

Nonparametric statistical analysis (e.g., determination of minimum, 25th percentile, median, 75th percentile, and maximum) was used for this study, as it is less sensitive to outliers and can handle non-detects. Results were graphed in a box-and-whisker plot, which summarized key nonparametric statistics and allowed for visual observations of similarities or differences in datasets. Moreover, the concentrations of carbamazepine and caffeine were plotted against that of primidone to identify possible correlations, and were further sorted and analyzed – along with NDMA precursor concentrations – based on low-primidone, middle-primidone, and high-primidone tertiles.

RESULTS AND DISCUSSION

Wastewater Impact

The results from 2006 are summarized according to different watersheds (states sampled) in Table 3. In 2007 (data not shown), additional sampling was conducted in other states (i.e., Michigan, Nevada, and Texas). The levels [10th percentile – maximum (median)] of primidone, carbamazepine, caffeine, and NDMAFP at all 20 sites (13 sites in 2006; in 2007, nine sites were re-sampled, plus an additional seven sites not included in 2006 were sampled) were 2-95 (7) ng/l, 2-207 (18) ng/l, 7-687 (78) ng/l, and 12-321 (35) ng/l, respectively. Because the scope of work of the overall project focused on the occurrence and formation of N-DBPs at DWTPs, WWTP effluents

were not sampled in this study except for two, resulting in 66-95 ng/l for primidone, 188-207 ng/l for carbamazepine, 48-202 ng/l for caffeine, and 321 ng/l for NDMAFP (only measured at one WWTP). In a previous study, primidone concentrations in WWTP effluents (some of which were in the watersheds in the current study) typically ranged from 100 to 200 ng/l (Krasner *et al.*, 2006), whereas 25th to 75th percentile NDMAFPs in WWTP effluents were 424-1,050 ng/l (Krasner *et al.*, 2008).

In one watershed in Pennsylvania (PA1) in 2006, primidone was detected at 5 ng/l at River B upstream of DWTP 1, and increased to 10 ng/l at DWTP 1. The 100% increase in concentration was five times the precision of the analytical method, which suggests that treated wastewater discharges in a tributary that entered River B near the intake for DWTP 1 affected this plant (Figure 2). Although the WWTP discharge in this tributary was not sampled in 2006, it was sampled three times in 2004-2005, and the concentration of primidone was 117-230 ng/l (Krasner *et al.*, 2008). During the latter sample events, primidone at DWTP 1 was 12-23 ng/l, where the amount detected at the DWTP was 10-11% of the concentration in the upstream WWTP discharge. These historical data gave an order of magnitude estimate of the degree to which this river may be effluent impacted. The concentration of primidone at DWTP 2 was similar (4 ng/l) to that of River B upstream of DWTP 1, indicating that the wastewater-impacted tributary did not appear to impact DWTP 2 (DWTP 2 is on the other side of the river). These data were consistent with the NDMAFP results from the corresponding sites, as the concentrations of the three PPCPs and NDMAFP were the highest at DWTP 1. The concentrations of caffeine in

TABLE 3. Occurrence (ng/l) of Three PPCPs and NDMAFP in Six Watersheds Sampled in 2006.

State	Site Description	Primidone	Carbamazepine	Caffeine	NDMAFP
CA	WWTP effluent	66	188	202	NA
	River A downstream of other WWTPs	62	125	26	NA
PA1	River B upstream of DWTP 1	5	18	133	17
	DWTP 1 influent	10	26	140	37
PA2	DWTP 2 influent	4	18	130	27
	River C upstream of DWTP 3	6	14	86	36
OK	DWTP 3 influent	6	16	78	35
	DWTP 4 influent	28	48	55	22
NJ	DWTP 5 influent	35	156	99	75
CO	River D upstream of DWTP 6	37	116	687	261
	River E upstream of DWTP 6	8	22	176	94
	DWTP 6 influent	9	5	207	53
	DWTP 7 influent	2	2	7	11

Notes: DWTP, drinking water treatment plant; NA, not analyzed; NDMAFP, formation potential for *N*-nitrosodimethylamine; PPCP, pharmaceutical and personal care product; WWTP, wastewater treatment plant.

DWTP 4 used water from an effluent-impacted lake; DWTP 7 influent was upstream of the metropolitan urban area.

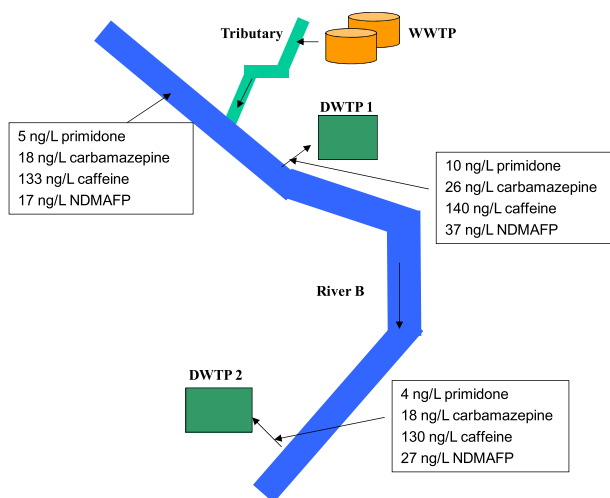


FIGURE 2. Sampling Locations and Concentrations of the Three PPCPs and NDMAFP in Watershed PA1 in 2006 (distance between DWTP 1 and DWTP 2 is ~3 miles).

this watershed, however, were not significantly different, with a relative standard deviation of 4% for the three sites sampled.

In a watershed in Colorado in 2006 (Figure 3), substantially higher levels of indicators (37 ng/l of primidone, 116 ng/l of carbamazepine, and 687 ng/l of caffeine) and NDMAFP (261 ng/l) were detected at River D upstream of DWTP 6. This watershed is considered to be a highly effluent-impacted stream, with discharges from WWTPs providing 10% to more than 50% of the river flow most of the year (Krasner *et al.*, 2006). Although the WWTP discharge in this river was not sampled in 2006, it was sampled three times in 2004-2005, and the concentration of primidone was 96-266 ng/l (Krasner *et al.*, 2008). During the latter sample events, primidone was not detected in River D upstream of WWTP discharges, whereas it was found in the river downstream of WWTPs and upstream of DWTP 6 (i.e., 22-68 ng/l), where the amount detected in the river was 23-28% (average = 26%) of the concentration in the upstream WWTP discharge. During the latter sample events, the discharges from upstream WWTPs were 25-27 million gallons per day (mgd) and the flow in the river downstream of the WWTPs and upstream of the DWTP was 53-184 mgd, where the contribution of treated wastewater discharges to the river flow was 14-50% (average = 32%). These historical data gave an order of magnitude estimate of the degree to which this river may be effluent impacted. Another sample site at River E upstream of DWTP 6 had lower levels of indicators (8 ng/l primidone, 22 ng/l carbamazepine, and 176 ng/l caffeine) and NDMAFP (94 ng/l), suggesting less impact from treated waste-

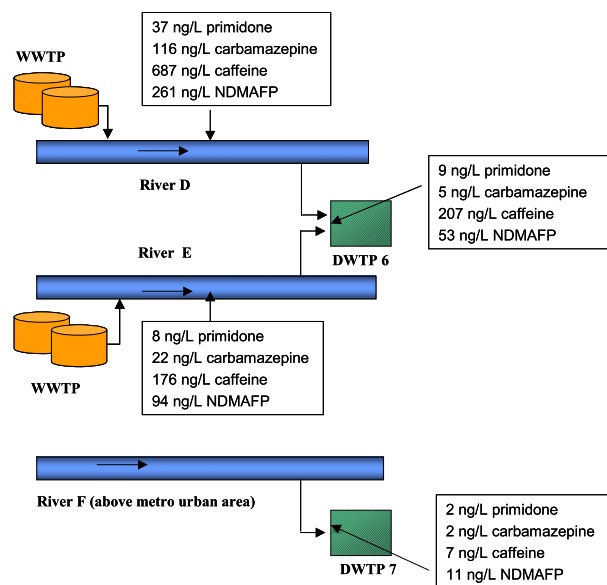


FIGURE 3. Sampling Locations and Concentrations of the Three PPCPs and NDMAFP in Colorado Watershed in 2006 (distance between WWTP on River D and DWTP 6 is ~17 miles).

water. The concentrations of the three PPCPs in River D were four to five times higher than in River E and the NDMAFP was three times higher. DWTP 7 took water from River F in the same watershed, but upstream of the metropolitan urban area. This DWTP had the lowest levels of the three PPCPs and NDMAFP of all samples collected in 2006, suggesting that the wastewater impact was minimal.

Excluding the two WWTP effluents, the other samples represent streams and DWTP intakes with various degrees of treated wastewater impact. The distributions of the three PPCPs and NDMAFP in these effluent-impacted surface waters sampled in 2006 and 2007 (W. A. Mitch *et al.*, 2008) are shown in box-and-whisker plots below (Figure 4).

Temporal Variations

Temporal variations in the occurrence of the three PPCPs and NDMAFP in two watersheds (PA1 and Colorado) are shown in Tables 4 and 5. In addition, streamflow data are provided, where available.

For watershed PA1, the changes in the concentrations of the three PPCPs and NDMAFP at DWTP 1 and DWTP 2 were minimal between the two sampling events in August 2006 and April 2007. The intention of the sampling in this watershed was to capture the occurrence data representing one dry season and one wet season. However, the streamflow (measured at River B) was actually unchanged and,

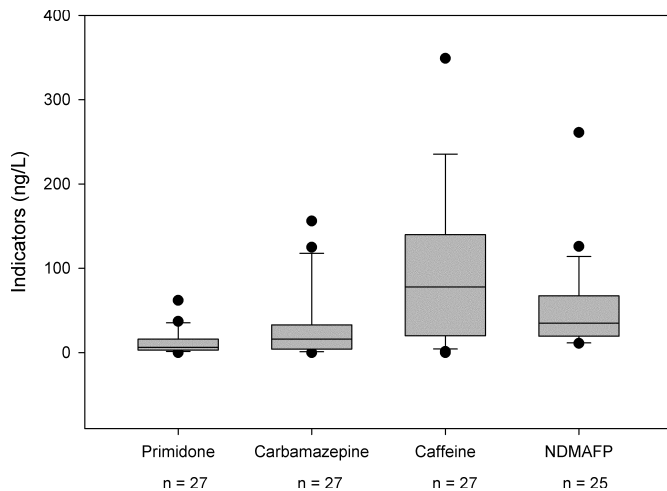


FIGURE 4. Concentrations of the Three PPCPs and NDMA Precursors in Effluent-Impacted Surface Waters Sampled in 2006 and 2007 [y-axis cut off at 400 ng/L; however, there was one caffeine result at a higher value (i.e., 687 ng/L)]. Top and bottom of box = 75th and 25th percentiles, respectively; top and bottom of whiskers = 90th and 10th percentiles, respectively; line across inside of box = median (50th percentile); and points beyond whiskers = outliers. Source: Reproduced from W.A. Mitch *et al.* (2008). Copyright Awwa Research Foundation.

therefore, the dilution of the wastewater discharges in this watershed stayed the same, resulting in similar concentrations of the analytes in the DWTP influents between the two sampling events. At the tributary site, the flow was ~100 cfs and was much

lower than that in the river (~2,600 cfs), resulting in less dilution of the WWTP effluent and, hence, higher concentrations of primidone (5.6×), carbamazepine (3.8×), and NDMAFP (6.2×) than in the samples collected in the mainstream of the river upstream of the tributary impacted by wastewater effluent. The caffeine concentration in this watershed, however, stayed at similar levels during both sampling events and at all sites sampled.

As for the Colorado watershed, the concentrations of the three PPCPs and NDMAFP were each approximately twice as high (1.5× to 2.3×) at the River D site in 2006 than in 2007, even though samples were both collected in September. This can be explained by year 2006 being a relatively dry year with streamflow (measured at River D) only half of the flow measured in year 2007, resulting in less dilution of WWTP outflows. Likewise, the concentrations of the three PPCPs were higher (1.8× to 2.9×) in River E in 2006 when compared with 2007, although the NDMAFP was not substantially higher (1.2×), where streamflow was much higher (3.4×) in 2007.

Correlation Analysis

Because primidone had been demonstrated previously (Krasner *et al.*, 2006) to be a conservative indicator of treated wastewater impact, the concentrations of carbamazepine and caffeine were plotted

TABLE 4. Temporal Variations in Watershed PA1.

Site Location	Primidone (ng/l)		Carbamazepine (ng/l)		Caffeine (ng/l)		NDMAFP (ng/l)		Streamflow (cfs)	
	2006-08-08	2007-04-12	2006-08-08	2007-04-12	2006-08-08	2007-04-12	2006-08-08	2007-04-12	2006-08-08	2007-04-12
River B	5	-	18	-	133	-	17	-	2,610	~2,600 ¹
Tributary	-	28	-	68	-	138	-	106	-	~100
DWTP 1	10	11	26	33	140	142	37	50	-	-
DWTP 2	4	8	18	19	130	125	27	28	-	-

Notes: DWTP, drinking water treatment plant; NDMAFP, formation potential for *N*-nitrosodimethylamine.

Dates are in YYYY-MM-DD notation.

¹Estimated.

TABLE 5. Temporal Variations in Colorado Watershed.

Site Location	Primidone (ng/l)		Carbamazepine (ng/l)		Caffeine (ng/l)		NDMAFP (ng/l)		Streamflow (cfs)	
	2006-09-12	2007-09-13	2006-09-12	2007-09-13	2006-09-12	2007-09-13	2006-09-12	2007-09-13	2006-09-12	2007-09-13
River D	37	24	116	50	687	349	261	126	110	226
River E	8	3	22	12	176	60	94	76	21	72
DWTP 6	9	16	5	13	207	185	53	26	-	-
DWTP 7	2	ND	2	1	7	12	11	12	-	-

Notes: DWTP, drinking water treatment plant; ND, not detected; NDMAFP, formation potential for *N*-nitrosodimethylamine.

Dates are in YYYY-MM-DD notation.

against that of primidone to identify any correlations among the three PPCPs in samples collected from all the watersheds in 2006 and 2007 (Figure 5). The correlation between the two anticonvulsants, carbamazepine and primidone, was good ($R^2 = 0.90$), whereas there was no correlation between caffeine and primidone. The lack of correlation between caffeine and other pharmaceuticals has also been noted by Glassmeyer *et al.* (2005).

To further examine the correlations among the three PPCPs and between primidone and NDMAFP in wastewater-impacted waters, the datasets – excluding the two WWTP effluents collected in 2006 and 2007 and the wastewater-dominated stream sample in California in 2006 (Table 3) – were sorted in ascending order based on the primidone results and then were divided into three groups (tertiles) of eight to nine sets of data. The lowest tertile with eight sets of data had ND (<1)–3.9 ng/l of primidone, the middle tertile with nine sets of data had 4–8 ng/l, and the highest tertile with nine sets of data had 9–37 ng/l. The low-tertile results came from those sites minimally impacted by WWTP discharges, whereas the high-tertile results were from the most wastewater-impacted sites in this study. Figures 6 to 8 show the occurrence of the other two PPCPs and NDMA precursors in each tertile (W.A. Mitch *et al.*, 2008). In the case of the low-primidone tertiles, only the boxes (75th and 25th percentiles and medians) are shown as there were inadequate sets of data to compute the whiskers (90th and 10th percentiles). For the low-primidone, moderate-primidone, and high-primidone tertiles, carbamazepine was ND (<1)–12 (median = 4), ND–27 (median = 18), and 5–156 (median = 48) ng/l,

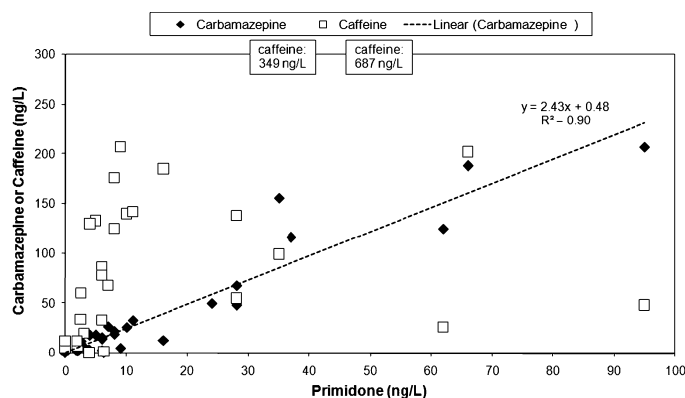


FIGURE 5. Correlations of Concentrations of Carbamazepine and Caffeine to That of Primidone in All the Watersheds in 2006 and 2007 (y-axis cut off at 300 ng/l; however, there were two caffeine results at higher values as shown). Source: Reproduced from W.A. Mitch *et al.* (2008). Copyright Awwa Research Foundation.

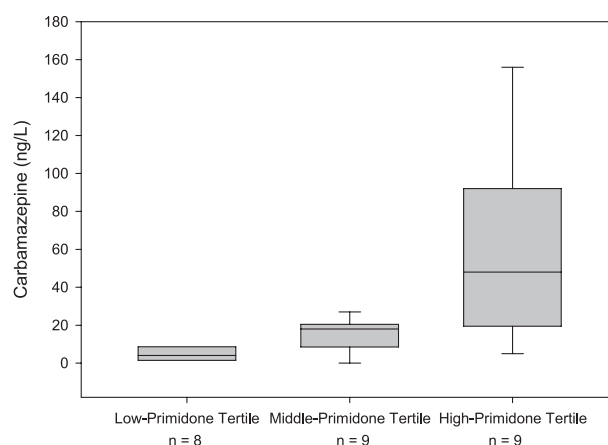


FIGURE 6. Occurrence of Carbamazepine in Wastewater-Impacted Samples Sorted by Primidone Tertiles. For the low-primidone tertile, the whiskers (90th and 10th percentiles) were not computed because n was less than 9 (insufficient data to compute those statistics). Source: Reproduced from W.A. Mitch *et al.* (2008). Copyright Awwa Research Foundation.

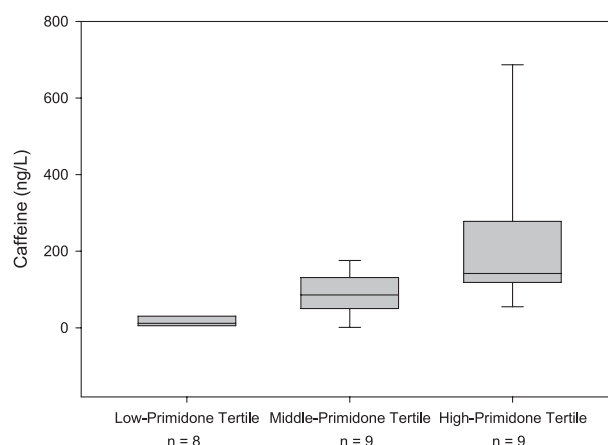


FIGURE 7. Occurrence of Caffeine in Wastewater-Impacted Samples Sorted by Primidone Tertiles. Source: Reproduced from W.A. Mitch *et al.* (2008). Copyright Awwa Research Foundation.

respectively. For caffeine, the three groups were ND (<1)–60 (median = 12), 1–176 (median = 86), and 55–687 (median = 142) ng/l, respectively. So on a central tendency basis, increased levels of primidone were associated with increased levels of other PPCPs in this study. In the low-primidone group, NDMAFP ranged from 10 to 76 ng/l (median = 20 ng/l), whereas in the middle-primidone and high-primidone tertiles, NDMAFP was 11–96 ng/l (median = 28 ng/l) and 22–261 ng/l (median = 53 ng/l), respectively. On a central tendency basis, the presence of NDMA precursors was higher in the wastewater-impacted waters. Although the presence of high NDMAFP suggests wastewater impact, it is not considered a

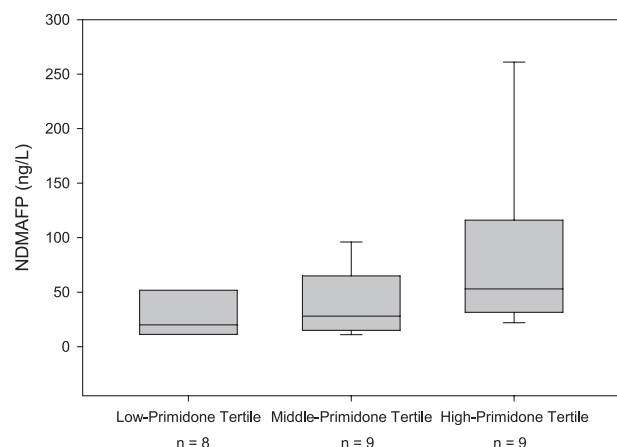


FIGURE 8. Occurrence of NDMA Precursors in Wastewater-Impacted Samples Sorted by Primidone Tertiles. Source: Reproduced from W.A. Mitch *et al.* (2008). Copyright Awwa Research Foundation.

quantitative indicator, as its concentrations in different WWTP discharge can vary greatly from <100 to >1,000 ng/l (Krasner *et al.*, 2008). In contrast, the primidone concentrations in WWTP discharge were generally in the 100-200 ng/l range (Krasner *et al.*, 2006), and did not vary much at different geographical locations, which indicates that primidone is a useful quantitative indicator of wastewater impact.

CONCLUSIONS

The three PPCPs and NDMA precursors were present in most of the wastewater-impacted samples. The correlation between the concentrations of carbamazepine and primidone – which was shown in previous research as a quantitative indicator of treated wastewater impact – suggests that either pharmaceutical can be used as a conservative wastewater indicator, whereas the concentration of caffeine did not appear to provide the same degree of information. The presence of NDMAFP was substantially higher in the effluent-impacted waters with higher levels of primidone; however, it is not a quantitative indicator of wastewater impact. In two case studies, temporal variations in the concentrations of the three PPCPs (with the exception of caffeine in samples from some sites) and NDMAFP reflected changes in streamflow between sampling events. Based on the limited data from this project, the measurement of certain PPCPs and NDMAFP in drinking water sources can be used in evaluating wastewater impact in watersheds.

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